

BEE 4750 Homework 2: Systems Modeling and Simulation

Name:

ID:

Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `c:\Users\maozb\Documents\Cornell\BEE4750\Hw2\hw2-mb`
Installed JpegTurbo_jll          v3.1.3+0
Installed Libmount_jll          v2.41.1+0
Installed Accessors              v0.1.42
Installed InvertedIndices        v1.3.1
Installed InlineStrings          v1.4.5
Installed Preferences            v1.5.0
Installed Roots                  v2.2.10
Installed DataFrames             v1.8.0
Installed Fontconfig_jll        v2.17.1+0
Installed WorkerUtilities        v1.6.1
Installed WeakRefStrings         v1.4.2
Installed SentinelArrays         v1.4.8
Installed DataStructures         v0.19.1
Installed StatsBase              v0.34.6
Installed Expat_jll             v2.7.1+0
Installed PooledArrays           v1.4.3
Installed TableTraits            v1.0.1
Installed DataValueInterfaces    v1.0.0
Installed ConstructionBase       v1.6.0
Installed Plots                  v1.40.19
Installed PrettyTables           v3.0.8
Installed Ghostscript_jll        v9.55.1+0
Installed CompositionsBase       v0.1.2
Installed Libuuid_jll            v2.41.1+0
Installed CommonSolve            v0.2.4
Installed InverseFunctions        v0.1.17
Installed OpenSSL_jll            v3.5.2+0
Installed Crayons                 v4.1.1
Installed Latexify                v0.16.10
Installed Tables                  v1.12.1
Installed IteratorInterfaceExtensions v1.0.0
Installed Glib_jll               v2.86.0+0
Installed FilePathsBase          v0.9.24
Installed CSV                     v0.10.15
Installed StringManipulation      v0.4.1
Installed SortingAlgorithms       v1.2.2
Precompiling project...
1217.0 ms   InverseFunctions
1239.5 ms   WorkerUtilities
1408.8 ms   IteratorInterfaceExtensions
1186.5 ms   CompositionsBase
1224.2 ms   CommonSolve

```

1214.6 ms	DataValueInterfaces
1203.8 ms	InlineStrings
754.6 ms	InvertedIndices
888.0 ms	ConstructionBase
1202.5 ms	PooledArrays
1363.0 ms	Preferences
2004.6 ms	Crayons
912.1 ms	InverseFunctions → InverseFunctionsDatesExt
2213.8 ms	SentinelArrays
923.0 ms	LogExpFunctions → LogExpFunctionsInverseFunctionsExt
1598.0 ms	InverseFunctions → InverseFunctionsTestExt
1971.2 ms	FilePathsBase
2882.9 ms	DataStructures
760.0 ms	TableTraits
860.5 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
779.3 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
1422.7 ms	Unitful → InverseFunctionsUnitfulExt
952.9 ms	JLLWrappers
2176.8 ms	SortingAlgorithms
2375.7 ms	FilePathsBase → FilePathsBaseMmapExt
2822.7 ms	Unitful → ConstructionBaseUnitfulExt
2390.0 ms	PrecompileTools
2198.3 ms	FilePathsBase → FilePathsBaseTestExt
1873.7 ms	Tables
961.6 ms	EpollShim_jll
979.5 ms	Libmount_jll
1118.6 ms	Graphite2_jll
1112.7 ms	Bzip2_jll
1160.8 ms	LLVMOpenMP_jll
1201.6 ms	OpenSSL_jll
4100.3 ms	Accessors
1015.4 ms	Xorg_libICE_jll
1000.3 ms	Xorg_libXau_jll
1156.1 ms	libfdk_aac_jll
1166.7 ms	libpng_jll
1150.8 ms	LAME_jll
1143.8 ms	LERC_jll
980.7 ms	fzf_jll
1119.9 ms	XZ_jll
1131.7 ms	JpegTurbo_jll
956.3 ms	mtdev_jll
979.9 ms	Xorg_libXdmcp_jll
1005.9 ms	Ogg_jll

1055.4 ms	x265_jll
990.4 ms	x264_jll
1109.9 ms	Zstd_jll
1123.1 ms	libaom_jll
1099.9 ms	LZO_jll
1024.5 ms	Xorg_xtrans_jll
1117.6 ms	Expat_jll
1183.4 ms	Opus_jll
1065.0 ms	libevdev_jll
1050.1 ms	Libiconv_jll
1042.3 ms	Libffi_jll
949.9 ms	Libuuid_jll
966.8 ms	eudev_jll
1066.3 ms	FriBidi_jll
1138.4 ms	FreeType2_jll
1111.6 ms	Pixman_jll
2520.2 ms	RecipesBase
1269.9 ms	Accessors → LinearAlgebraExt
2728.2 ms	StringManipulation
1379.0 ms	Accessors → TestExt
4102.0 ms	StatsBase
951.1 ms	Xorg_libSM_jll
2476.0 ms	OpenSSL
2060.1 ms	Accessors → UnitfulExt
1027.9 ms	Xorg_libxcb_jll
1687.7 ms	JLFzf
1722.6 ms	Ghostscript_jll
1818.5 ms	libvorbis_jll
876.8 ms	DBus_jll
5556.4 ms	ColorSchemes
912.9 ms	Wayland_jll
1539.7 ms	Libtiff_jll
1372.7 ms	GettextRuntime_jll
908.0 ms	libinput_jll
886.9 ms	Xorg_xcb_util_jll
1827.4 ms	Fontconfig_jll
1010.2 ms	Xorg_libX11_jll
1651.2 ms	Glib_jll
871.9 ms	Xorg_xcb_util_image_jll
4549.3 ms	Roots
879.5 ms	Xorg_xcb_util_keysyms_jll
977.9 ms	Xorg_xcb_util_renderutil_jll
4350.9 ms	Latexify

1005.6 ms	Xorg_xcb_util_wm_jll
939.6 ms	Xorg_libXrender_jll
966.1 ms	Xorg_libXext_jll
1046.1 ms	Xorg_libXfixes_jll
1441.6 ms	Xorg_libxkbfile_jll
1670.8 ms	Xorg_xcb_util_cursor_jll
2180.5 ms	Roots → RootsUnitfulExt
1114.1 ms	Xorg_libXinerama_jll
15925.4 ms	Parsers
1718.6 ms	Latexify → SparseArraysExt
2979.6 ms	UnitfulLatexify
946.4 ms	Xorg_libXrandr_jll
964.9 ms	Libglvnd_jll
919.1 ms	Xorg_libXcursor_jll
983.7 ms	Xorg_libXi_jll
979.1 ms	Xorg_xkbcomp_jll
1881.4 ms	Cairo_jll
940.0 ms	InlineStrings → ParsersExt
922.6 ms	Xorg_xkeyboard_config_jll
940.8 ms	xkbcommon_jll
2419.7 ms	JSON
1465.6 ms	WeakRefStrings
2192.9 ms	HarfBuzz_jll
1284.7 ms	Vulkan_Loader_jll
15180.2 ms	PlotUtils
2692.8 ms	libass_jll
2610.9 ms	Pango_jll
2284.7 ms	Qt6Base_jll
885.5 ms	libdecor_jll
3443.0 ms	FFMPEG_jll
5095.4 ms	PlotThemes
1768.0 ms	GLFW_jll
6942.6 ms	Qt6ShaderTools_jll
7999.7 ms	RecipesPipeline
25397.9 ms	HTTP
3810.0 ms	FFMPEG
3500.6 ms	GR_jll
3284.4 ms	Qt6Declarative_jll
715.9 ms	Qt6Wayland_jll
7070.9 ms	GR
22810.6 ms	CSV
38017.3 ms	PrettyTables
45486.7 ms	DataFrames

```
2605.8 ms    Latexify → DataFramesExt
72621.7 ms   Plots
4762.4 ms    Plots → UnitfulExt
137 dependencies successfully precompiled in 144 seconds. 82 already precompiled.
```

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of 2.3 kg/(1000m³). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance which includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1 - \alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m²/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m² but during the Neoproterozoic Era had a value of 1272 W/m² (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m²/°C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m² (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

```
function integrate_climate(T0, alpha, S, A, B, C, delta_t, N)
    # Initialize array to store temperatures
    T = zeros(Float64, N+1)
    T[1] = T0

    # Time stepping loop
    for n in 1:N
        dTdt = ( (1 - alpha)*S/4 - (A - B*T[n]) ) / C
        T[n+1] = T[n] + delta_t * dTdt
    end

    return T
end

C = 51.0 # J/m^2/°C
alpha = 0.3 # albedo
S = 1272.0 # W/m^2, solar constant for Snowball Earth
A = 221.2 # W/m^2
B = -1.3 # W/m^2/°C
delta_t = 1.0 # yr
T0 = 14 # initial temperature
N = 10 # simulate 100 years

# Run integration
temperature_series = integrate_climate(T0, alpha, S, A, B, C, delta_t, N)
```

```
11-element Vector{Float64}:
 14.0
13.670588235294119
13.349573241061131
13.03674098197526
12.731882878513145
12.434795667884378
12.145281268506933
```


11.863146647937148
11.588203694166202
11.32026909019726
11.059164191819683

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
# Ice-albedo feedback function
function albedo(T)
    _i = 0.5
    _0 = 0.3
    if T <= -10
        return _i
    elseif T >= 10
        return _0
    else
        return _i + (_0 - _i) * ((T + 10) / 20)
    end
end

# Changing the forward Euler integration function
function integrate_climate_feedback(T0, S, A, B, C, delta_t, N)
    T = zeros(Float64, N+1)
    T[1] = T0

    for n in 1:N
        alpha = albedo(T[n])
        dTdt = ( (1 - alpha)*S/4 - (A - B*T[n]) ) / C
        T[n+1] = T[n] + delta_t * dTdt
    end
end
```

```

    return T
end

# Parameters
C = 51.0
S = 1272.0
A = 221.2
B = -1.3
delta_t = 1.0
N = 300

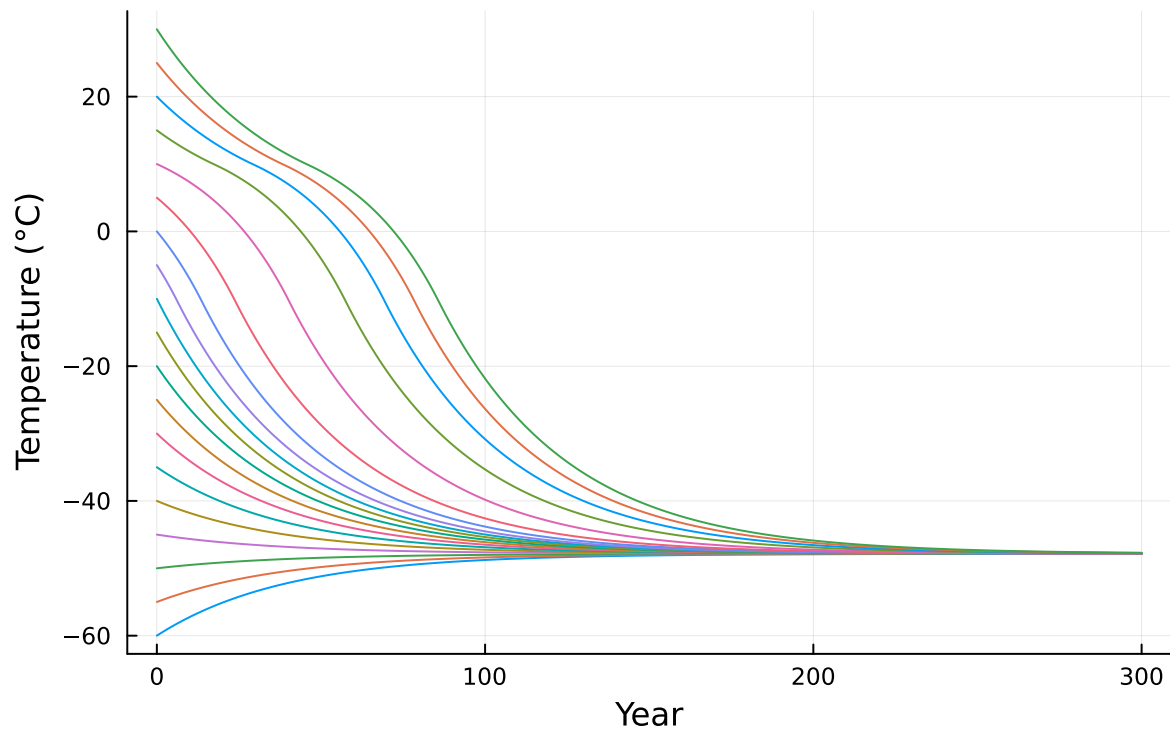
# Initial temperatures
T_initial = -60:5:30 # every 5°C from -60 to 30
trajectories = []

# Run simulation for each initial temperature
for T0 in T_initial
    push!(trajectories, integrate_climate_feedback(T0, S, A, B, C, delta_t, N))
end

# Plot all trajectories
plot()
for i in 1:length(T_initial)
    plot!(0:N, trajectories[i], label=false)
end
xlabel!("Year")
ylabel!("Temperature (°C)")
title!("Temperature trajectories with ice-albedo feedback")

```

Temperature trajectories with ice-albedo feedback



Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C ?

```
S = 1368
# Initial temperatures
T_initial = -60:5:30 # every 5°C from -60 to 30
trajectories = []

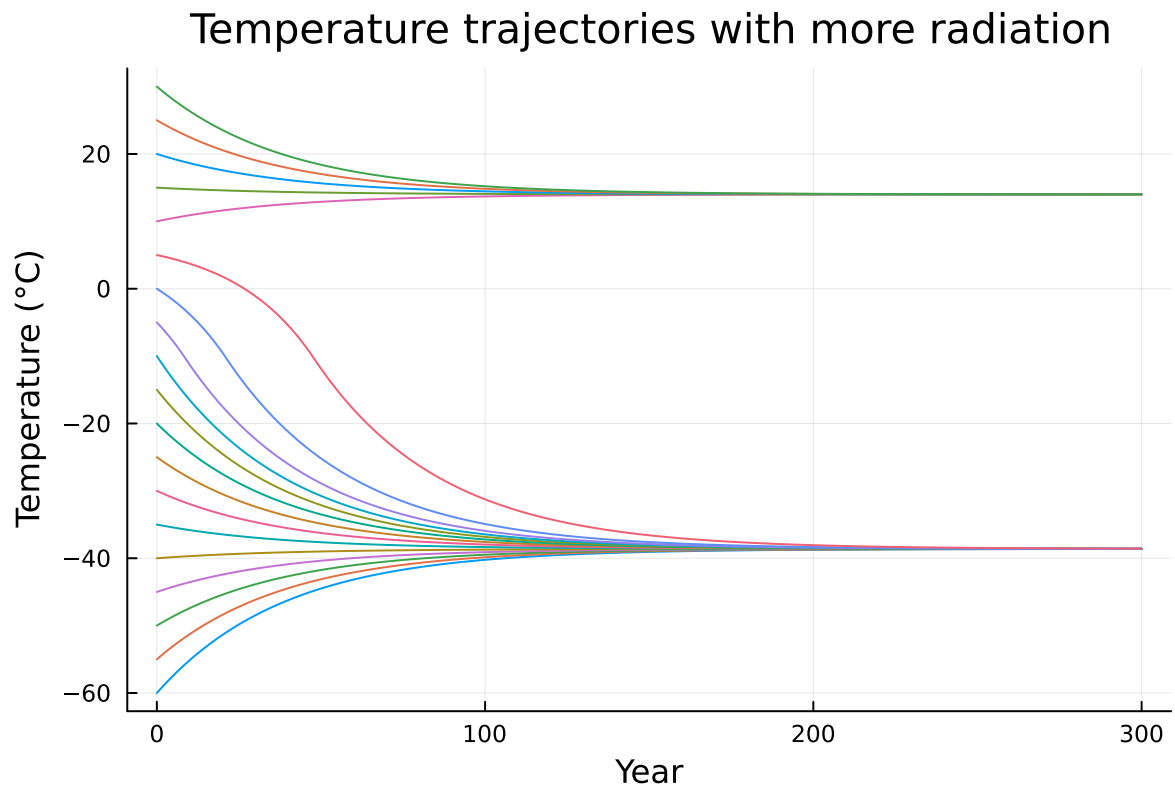
# Run simulation for each initial temperature
for T0 in T_initial
    push!(trajectories, integrate_climate_feedback(T0, S, A, B, C, delta_t, N))
end

# Plot all trajectories
plot()
```

```

for i in 1:length(T_initial)
    plot!(0:N, trajectories[i], label=false)
end
xlabel!("Year")
ylabel!("Temperature (°C)")
title!("Temperature trajectories with more radiation")

```



References

List any external references consulted, including classmates.